

Transportation Behavior and Safety Analysis

DS2202 – Data Integration and Warehousing

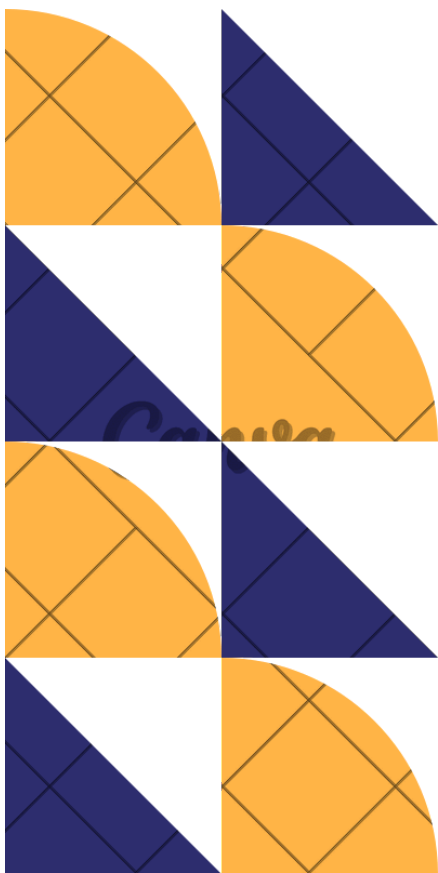
Group 5

Team Members:

- Maha Almjnoni – 446004071
- Dana Alqurashi – 446001073
- Rawan Alghamdi – 44610098

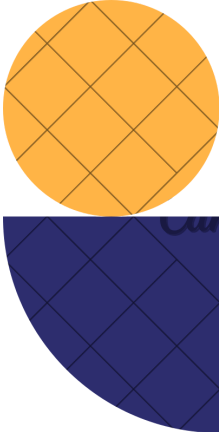
Project Scenario

Transportation Behavior and Safety Analysis This project aims to analyze people's travel behavior and study how it is related to traffic accidents, injuries, and deaths. The data will help in understanding patterns and improving transportation safety.





Lab 1 – Project Scenario & Data Sources

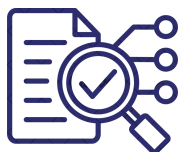


Project Scenario



Analyze people’s travel behavior and its relationship with traffic accidents, injuries, and fatalities to improve transportation safety.

Data Sources



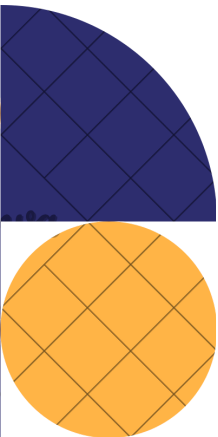
Source Name :	format :	records :
Travel Preference Survey	CSV	1000
Traffic Accidents HCM City	JSON	500
Injured and Dead in Accidents	Excel/CSV	1200



Goal:
Analyze the relationship between travel behavior and traffic accidents, injuries, and fatalities to improve transportation safety.



Key Outcome:
Selected heterogeneous datasets that can be integrated to support transportation safety analysis.





Lab 2 – Structured vs. Unstructured Data

Structured Data

- ID
- Keyword
- Location
- Target

Unstructured Data

- Text

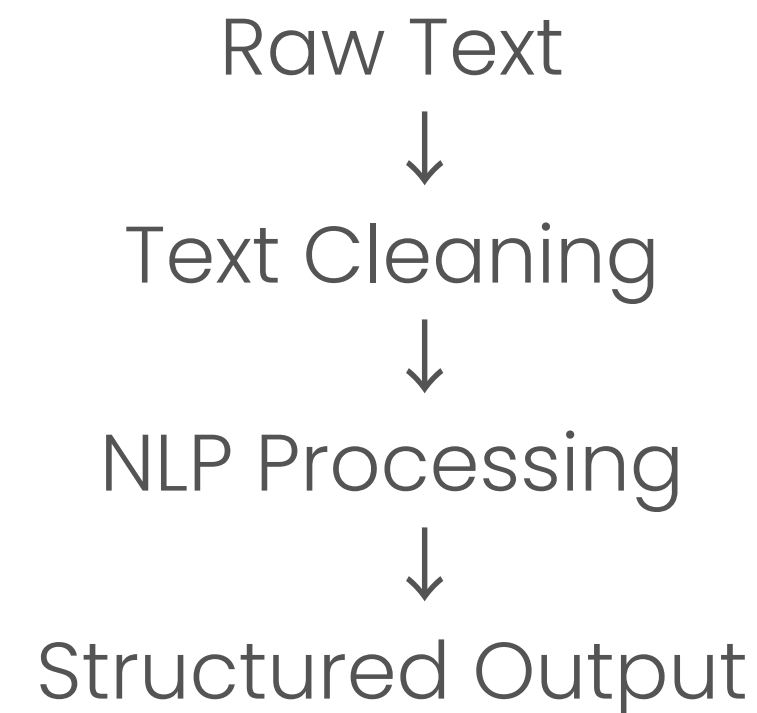
Transformation Process

- Text Cleaning
- Entity Extraction
- Sentiment Analysis
- Topic Extraction
- Word Count Calculation

Key Outcome:

Converted unstructured text into structured information suitable for analysis.

Attribute	Data Type	Structured / Unstructured	Justification
Id	integer	Structured	Unique identifier "primary key" - follows a strict format.
Keyword	string	Structured	Categorical value - limited set of possibilities.
Location	string	Structured	Geographical data - represents a known location
Text	string	Unstructured	Free text - contains slang, typos, URLs, and no fixed pattern. (The core unstructured element)
Target	integer	Structured	Binary label - a clean, organized classification (0 or 1)





Lab 3 – Data Cleaning & Preprocessing

Data Quality Issues

- Missing Values
- Duplicate Records
- Format Inconsistencies
- Invalid Entries
- Noisy Text Data

Cleaning Techniques

- Standardization
- Data Normalization
- Missing Value Handling
- Duplicate Removal
- Data Validation
- Text Preprocessing

Key Outcome:

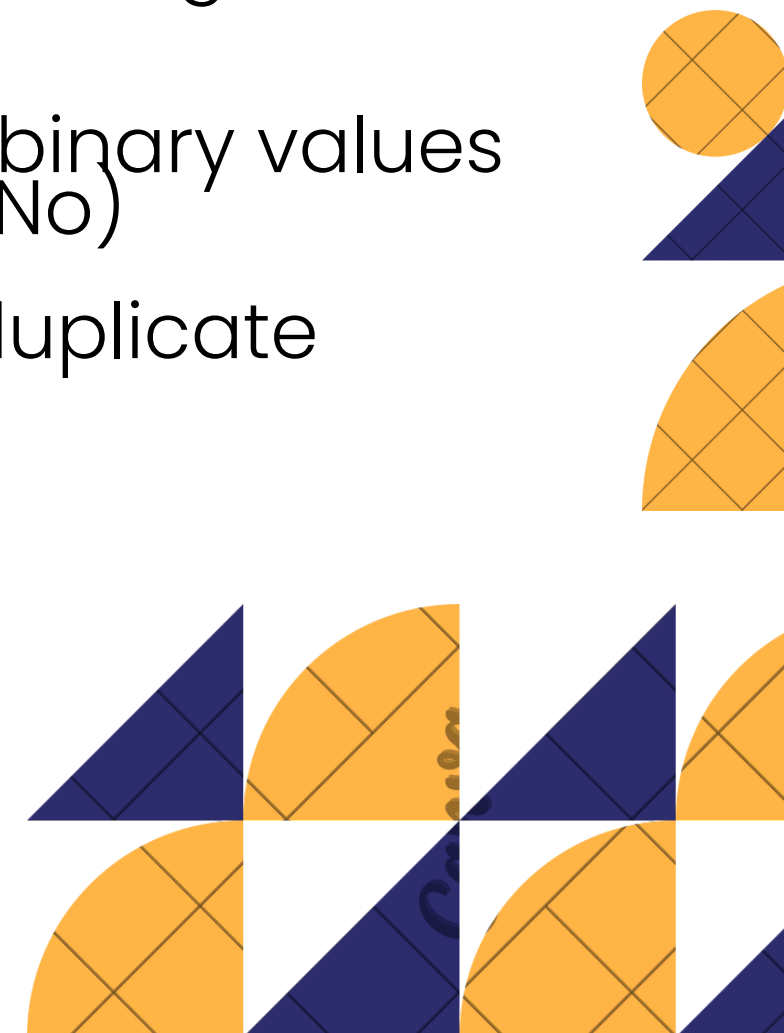
Improved data quality and prepared datasets for integration and

Issue	Cleaning Technique
Text format inconsistency in region_name	Standardization
Binary format (0/1) in injured_flag	Data transformation
Repeated values in region_number	Duplicate removal

Prepared clean and consistent data for Schema Matching and Data Integration

Examples of Cleaning Actions

- Standardized region names
- Converted binary values (0/1 → Yes/No)
- Removed duplicate records



Lab 4: Schema Matching

Identifying Correspondences Between Data Sources

Source A Attribute(Injured and Dead)	Source B Attribute(Traffic Accidents)	Match Type	Confidence
ID	accident_id	Match	High
gregorian_date	accident_time	Match	High
region_name	city	Match	High
accident_type	description	Partial Match	Medium

- **Matching Strategy:** Based on semantic meaning, data type compatibility, and attribute definitions.
- **Unmatched Attributes:** Features like male_count and female_count had no equivalents in Source B and were handled separately to maintain data quality.

Lab 5: Schema Mapping & Rules

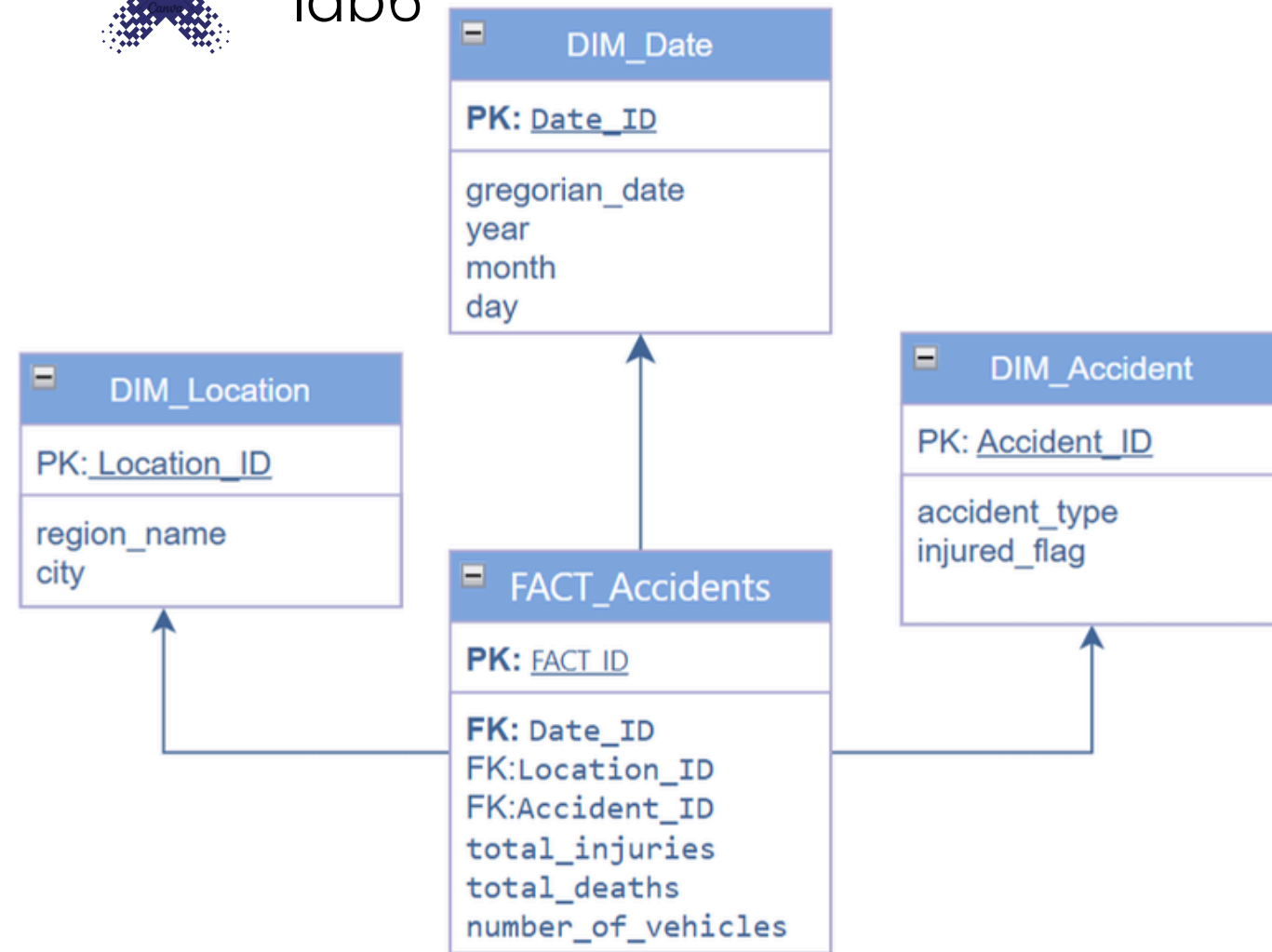
Logical Data Transformation Rules

Rule #	Rule Expression	Description
Rule 1	Target.accident_id = Source_A.ID	Maps the unique identifier (1:1 Mapping)
Rule 2	Target.accident_time = Source_A.gregorian_date	Maps the date and time directly (1:1 Mapping)
Rule 3	Target.description = Source_A.accident_type	Maps general categories into descriptive text (1:N Mapping)

- **Mapping Approach:** Conceptual and Logical mapping designed to define how data flows into the target schema without loss of information.
- **Implementation Tool:** Formulated logically as a baseline for the ETL process implemented via Python (Pandas) in the subsequent phases.



lab6



Tables Loaded:

- DIM_Date (10 rows)
- DIM_Location (10 rows)
- DIM_Accident (10 rows)
- Fact_Accidents (10 rows)

Date_ID	gregorian_date	year	month	day
1	2015-10-14	2015	10	14
2	2015-11-13	2015	11	13
3	2015-12-12	2015	12	12
4	2016-01-11	2016	1	11
5	2016-02-10	2016	2	10

Location_ID	region_name	city
1	Riyadh	Riyadh
2	Makkah	Jeddah
3	Eastern	Dammam
4	Madinah	Madinah
5	Qassim	Buraydah

ETL Process Summary



Extract 3 heterogeneous sources (CSV, JSON, Excel)

Transform Cleaning, normalization, type conversion

Load 10 rows inserted per table (sample)

DB TrafficDB – SQL Server



Lab7

1.Which city has the highest injuries?	Hail – 400 cases	Bar Chart SUM+ GROUP BY city
2.Which accident type causes the most deaths?	Death Accident – 45 cases	Pie Chart SUM + GROUP BY type
3.Which month has the highest accidents?	Month 7 (July) – 3 cases	Column Chart COUNT + GROUP BY month
4.Which city has most vehicles involved?	Madinah & Hail – 6 vehicles	Bar Chart SUM+ GROUP BY city

```
137 • SELECT
138     L.city,
139     SUM(F.total_injuries) AS Total_Injuries
140 FROM FACT_Accidents F
141 JOIN DIM_Location L
142 ON F.Location_ID = L.Location_ID
143 GROUP BY L.city
144 ORDER BY Total_Injuries DESC;
145
```

city	Total_Injuries
Hail	400
Madinah	339
Jeddah	338
Dammam	333
Buraydah	323
Arar	320
Riyadh	310
Abha	280
Jazan	210
Tabuk	190

```
146 • SELECT
147     A.accident_type,
148     SUM(F.total_deaths) AS Total_Deaths
149 FROM FACT_Accidents F
150 JOIN DIM_Accident A
151 ON F.Accident_ID = A.Accident_ID
152 GROUP BY A.accident_type
153 ORDER BY Total_Deaths DESC;
154
```

accident_type	Total_Deaths
Death Accident	45
Injury Accident	4

```
155 • SELECT
156     D.month,
157     COUNT(F.Fact_ID) AS Total_Accidents
158 FROM FACT_Accidents F
159 JOIN DIM_Date D
160 ON F.Date_ID = D.Date_ID
161 GROUP BY D.month
162 ORDER BY Total_Accidents DESC;
163
```

month	Total_Accidents
7	3
6	2
10	1
11	1
12	1
1	1
2	1

```
164 • SELECT
165     L.city,
166     SUM(F.number_of_vehicles) AS Total_Vehicles
167 FROM FACT_Accidents F
168 JOIN DIM_Location L
169 ON F.Location_ID = L.Location_ID
170 GROUP BY L.city;
```

city	Total_Vehicles
Riyadh	5
Jeddah	3
Dammam	4
Madinah	6
Buraydah	2
Abha	4
Tabuk	2
Arar	5
Jazan	3
Hail	6

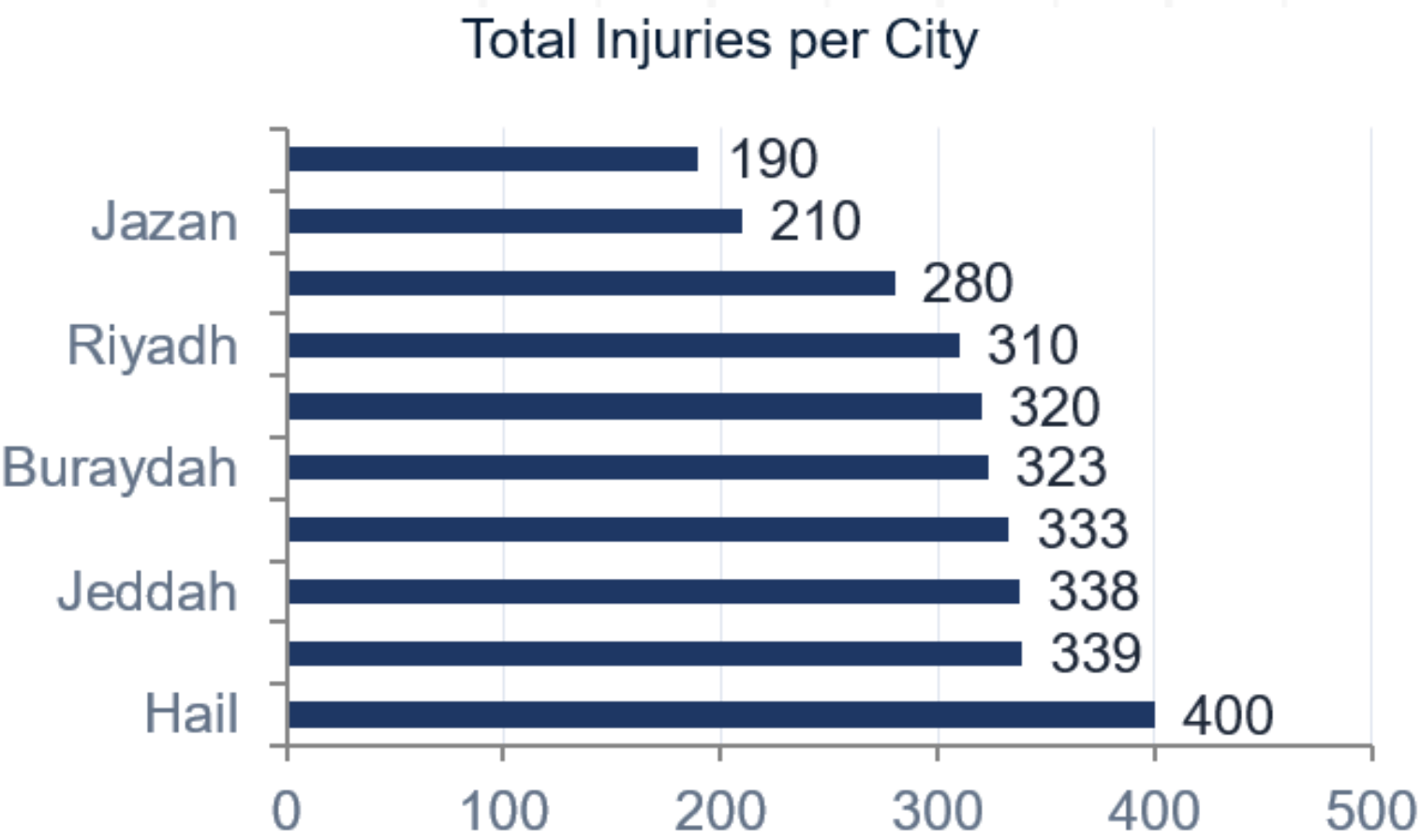
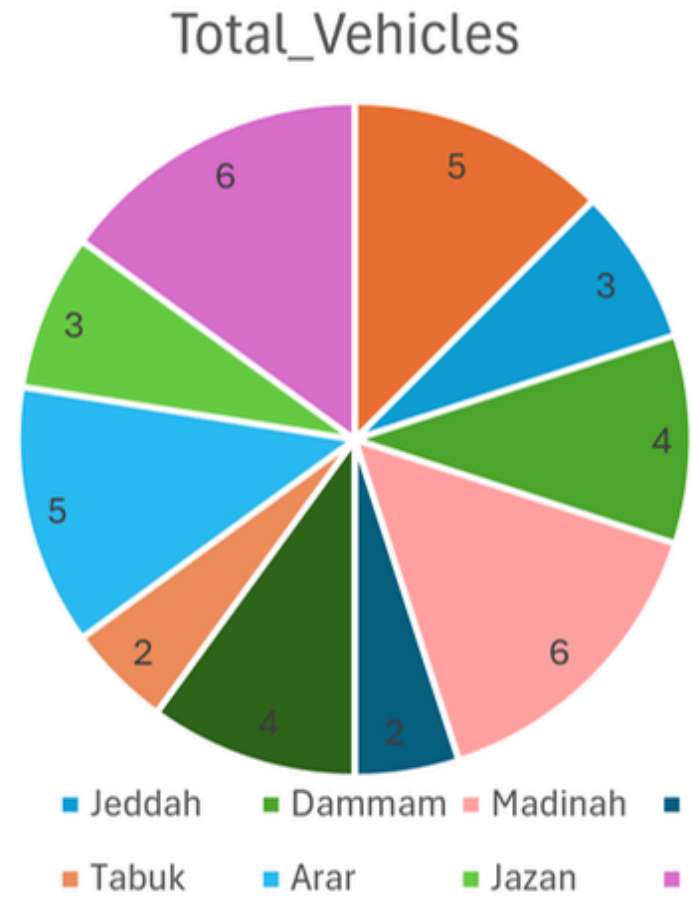
Query 1 – Total Injuries per City

Query 2 –Total Deaths per Accident Type

Query 3 –Accidents Trend by Month

Query 4 – Total Vehicles per City

Key Insights & Visual Dashboards



Recommendations

Based on the visual outputs and data analysis:

- Deploy more emergency services in Hail and Madinah due to high injury rates
- Launch targeted safety campaigns in July — peak accident month
- Implement specific prevention measures for Death Accident type — responsible for 92% of deaths
- Strengthen traffic management in cities with high multi-vehicle collisions

Lessons Learned

- Data cleaning is the foundation — quality in = quality out
 - Schema matching reveals hidden integration challenges early
 - Star schema simplifies complex multi-source querying
 - OLAP queries turn raw warehouse data into actionable decisions
 - Heterogeneous sources require careful format unification before integration
- 